WEEK -9

Minor Project (BTCS607N)-Weekly Progress Report

Project Title: Drone-Based Intelligent System for Apple Orchard Management in Himachal Pradesh.

Group ID: G9

Branch & Section: BTech CSE (AIML with Microsoft), AIDS -A

Date: 06-04-2026

1. Executive Summary

A. **Current Sprint:** Sprint 1 (Issue – 9)

B. **Sprint / Issue Goal:** To prepare the trained model and backend system for CI/CD implementation by ensuring proper dataset structuring, model training, and backend functionality.

C. **Status:** On Track

2. UML & OOAD Progress

A. **Diagrams Completed this week:** No new UML diagrams created

B. **Diagrams Revised:** No revisions done

C. **OO Concept Applied:**

Modular architecture
Separation of concerns (routes, services, utils)
Encapsulation of prediction and advisory logic

3.Technical Execution (DevOps & Git)

A. **Jira Tasks Completed:**

1. Dataset structuring for apple disease detection
2. Model training using YOLOv8
3. Model testing on sample images
4. Backend development using FastAPI
5. API endpoint /predict implementation
6. Integration of model with backend
7. Advisory system implementation

Link://ayushshrivastava.atlassian.net/jira/software/projects/DBISFAOM/boards/2?atlOrigin=eyJpIjoiY2I0Y2FjMmQ2MTliNDBlM2FjYzEyN2U1MjdkY2QzNGMiLCJwIjoiaiJ9

B. **Git Activity:**

1. Project pushed to GitHub repository
2. Backend and dataset structure organized
3. Model file (best.pt) added
4. Regular commits performed

Link : https://github.com/yp1818/Drone-Based-Apple-Orchard-Management

C. **CI/CD Status:**

1. CI/CD pipeline not implemented yet
2. Studied GitHub Actions workflow
3. Project prepared for future automation

4. System Architecture & Dataset Structure

1. Dataset Structure (YOLO Format)

```
dataset/
│
├── train/
├── validation/
├── test/
│
├── best.pt
├── data.yaml
```

Description:

1. Train, validation, and test folders used for proper data splitting
2. Ensures accurate training and evaluation
3. best.pt is the trained model
4. data.yaml defines class labels and dataset paths

2. Backend Project Structure

```
drone_orchard_system/
│
├── main.py
│
├── model/
│   └── best.pt
│
├── routes/
│   └── predict_route.py
│
├── services/
│   ├── prediction_service.py
│   └── advisory_service.py
│
├── utils/
│   └── file_handler.py
│
├── static/
│   └── uploads/
│
└── requirements.txt
```

Description:

1. main.py → Entry point of FastAPI server
2. model/ → Stores trained YOLO model
3. routes/ → API endpoints
4. services/ → Core logic (prediction + advisory)
5. utils/ → File handling
6. static/uploads/ → Stores uploaded images

FastAPI 0.1.0 OAS 3.1

/openapi.json

default

POST /predict/ Predict

Parameters

Cancel | Reset

No parameters

Request body required — multipart/form-data

file * required
string($binary) Choose File dataset-card.jpg

Execute | Clear

Responses

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/predict/' \
  -H 'accept: application/json' \
  -H 'Content-Type: multipart/form-data' \
  -F 'file=@dataset-card.jpg;type=image/jpeg'
```

Explain

Request URL

```
http://127.0.0.1:8000/predict/
```

Server response

Code	Details
200	Response body

Explain

```
{
  "result": {
    "disease": "black_rot",
    "confidence": 0.8893300294876099,
    "advisory": "Remove infected parts"
  }
}
```

Explain Download

Response headers

```
content-length: 101
content-type: application/json
date: Mon,06 Apr 2026 14:15:08 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	No links

Code	Description	Links
200	Successful Response	No links

Media type: application/json

Controls Accept header.

Example Value | Schema

```
"string"
```

422	Validation Error	No links

Media type: application/json

Example Value | Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string"
    }
  ]
}
```

Explain

GET / Home

4. Testing Summary

1. Model tested on multiple images
2. Verified predictions for all disease classes
3. Backend API tested successfully
4. Correct output generated:
5. Disease name
6. Confidence score
7. Advisory
8. System working without runtime errors

5. Goals for Next Week

1. Integrate the frontend with the backend API for predictions
2. Connect the system to a database to store prediction results and history
3. Update Jira tasks by moving them to “In Progress” and “Testing” stages

Mentor Signature **Project Guide Signature**